

Data Analytics Capstone

Using Machine Learning to Predict Complaint Resolution Outcomes in Retail Banking:

Evidence from Open Consumer Financial Complaint Data

Synopsis

Zahoor ul Islam

Walsh College

QM640: Data Analytics Capstone

Jainesh Garg

Summer 2026 Term

July 13, 2026

Introduction

Complaint management is a critical part of retail banking because complaints reveal moments where customer expectations, product performance, service delivery, and regulatory obligations meet. A complaint is not only an expression of dissatisfaction; it is also an operational signal. When complaint patterns are analyzed systematically, they may reveal recurring process failures, unclear product communication, weak digital journeys, delayed responses, or higher-risk service issues. Prior service recovery research shows that complaint handling influences customer satisfaction, trust, loyalty, and future intent (Homburg & Fürst, 2005; Maxham & Netemeyer, 2002; Tax et al., 1998).

In financial services, complaints also carry governance and conduct-risk implications. Retail banks operate in a trust-based environment where customers expect timely response, fair treatment, clear explanations, and reasonable remedies. Traditional complaint management commonly depends on manual categorization, first-in-first-out workflows, escalation rules, and post-resolution reporting. These methods help with compliance, but they are limited when complaint volumes are high and when the organization needs early warning signals before an issue becomes systemic.

The Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database provides an open source of structured and unstructured complaint data. The database includes variables such as product, sub-product, issue, sub-issue, company, state, submission channel, date received, date sent to the company, company response, timely response, and complaint narrative where consent is provided (Consumer Financial Protection Bureau [CFPB], 2025a, 2025b). The CFPB also provides programmatic access through its open data application

programming interface, allowing search results to be returned in formats such as JSON, CSV, XLS, and XLSX (CFPB Open Tech, 2025).

This capstone proposes an applied analytics study that uses open complaint data to predict complaint resolution outcomes in retail banking. The project is designed for a DBA-level audience because it combines managerial relevance, statistical hypothesis testing, machine learning, interpretability, and responsible AI considerations. The study is not intended to replace human judgment in complaint handling. Instead, it aims to test whether analytics can help complaint managers and customer-experience leaders identify risk patterns earlier and allocate attention more effectively.

Background and Context

Retail banking has become increasingly digital, but complaint handling remains a human-centered trust process. Customers may interact with a bank through mobile apps, cards, branches, call centers, payments, mortgages, or loans. When the customer journey fails, the complaint becomes evidence of a breakdown in service design, communication, back-office processing, or customer understanding. For banks, complaints matter because unresolved or poorly resolved cases can lead to attrition, negative word-of-mouth, regulatory attention, reputational damage, and higher operating cost.

Academic service literature emphasizes that customers evaluate complaint experiences through multiple justice dimensions. Distributive justice concerns whether the customer received a fair outcome. Procedural justice concerns whether the process was timely, consistent, and accessible. Interactional justice concerns whether the customer was treated respectfully and received adequate communication (Tax et al., 1998). These concepts are directly relevant to the CFPB data because the database captures product categories, issue categories, company response

types, response timing, and, in some cases, the consumer's own narrative description of what happened.

Machine learning can support complaint management by identifying combinations of complaint attributes associated with delayed responses or unfavorable outcomes. Logistic regression can provide an interpretable baseline, while tree-based methods such as random forest and gradient boosting can capture nonlinear relationships and interactions (Breiman, 2001; Chen & Guestrin, 2016; Friedman, 2001). Natural language processing can convert complaint narratives into quantitative features such as term frequency-inverse document frequency (TF-IDF), topic clusters, sentiment proxies, and text length. These features may improve prediction because narratives often contain details not captured by structured fields.

However, the use of complaint data also requires caution. CFPB notes that the complaint database is not a statistical sample of all consumer experiences and should not be interpreted as fully representative of all consumers, products, or companies (CFPB, 2025a). Company-level complaint volumes should also be viewed in the context of company size, market share, and geography. Therefore, this project will treat the CFPB dataset as a public complaint-risk signal rather than a complete measure of customer experience across the entire market.

Problem Statement

Retail banks often manage complaints reactively after customers have already experienced dissatisfaction, escalation, or service failure. Although complaint data contain valuable indicators of product, issue, channel, company, geography, and narrative risk, many organizations do not use these data to predict complaint resolution outcomes at intake or early in the workflow. This creates a gap between complaint reporting and proactive complaint-risk management. The proposed study addresses this gap by testing whether open consumer

complaint data can predict complaint resolution outcomes and identify the most significant factors associated with response timeliness and resolution type in retail banking.

Purpose of the Study

The purpose of this study is to develop and evaluate statistical and machine-learning models to predict complaint-resolution outcomes in retail banking using open CFPB consumer complaint data. The study will classify and explain outcomes such as timely response and company response type. It will also test whether complaint product, issue category, submission channel, company, geography, and complaint narratives are statistically significant predictors. The practical purpose is to generate evidence-based recommendations for complaint prioritization, customer experience improvement, operational-risk monitoring, and responsible AI governance.

Scope and objectives

The scope of this project is limited to open complaint data published by the CFPB. The proposed study will focus on retail banking-related complaint products from January 1, 2017, through December 31, 2025. The starting point is selected to reduce inconsistency from older complaint-category structures and to provide a multi-year dataset with sufficient volume. The project will exclude non-retail categories where the complaint relationship is less directly linked to retail banking service delivery. The final product list will be documented in the reproducible code repository.

Preliminary retail banking product categories will include checking or savings accounts, credit cards or prepaid cards, mortgages, vehicle loans or leases, personal loans, money transfers, and other consumer banking products where the complaint is directly connected to retail banking

service delivery. Debt collection records will be included only where the complaint can be reasonably linked to a consumer retail banking product. The final inclusion and exclusion list will be documented in the GitHub repository to ensure transparency and reproducibility.

The target outcomes are timely response and company response to the consumer. Timely response is useful because it is a clear binary field. Company response type is useful because it provides richer business interpretation, such as closed with explanation, closed with monetary relief, closed with non-monetary relief, or other response categories. Complaint narratives will be included only where consumers provided consent and the CFPB published the text after taking steps to remove personal information.

The objectives of the study are to:

- identify the complaint products, issue categories, channels, companies, and geographic factors associated with response timeliness;
- test whether issue categories significantly predict company response outcomes;
- evaluate whether NLP features from complaint narratives improve predictive performance over structured fields alone;
- compare interpretable statistical models with machine learning models for classification performance and business usefulness;
- assess the effect of class imbalance and compare imbalance-mitigation techniques for rare complaint-resolution outcomes;
- translate model findings into practical recommendations for complaint triage, customer experience improvement, and responsible AI controls.

Research Questions and Hypotheses

Table 1

Research Questions, Hypotheses, and Primary Methods

Research Question	Null Hypothesis (H0)	Alternative Hypothesis (H1)	Primary Method
RQ1: Does the complaint product type significantly influence the timeliness of the company's response?	Complaint product type has no statistically significant relationship with timely company response.	Complaint product type has a statistically significant relationship with the timeliness of company response.	Chi-square test; binary logistic regression
RQ2: Do complaint issue categories significantly predict company response outcome?	The complaint issue category does not significantly predict the company response outcome.	The complaint issue category significantly predicts the company response outcome.	Multinomial logistic regression; random forest
RQ3: Do complaint narratives improve prediction performance compared with structured fields only?	Narrative text features do not significantly improve model performance compared with structured variables only.	Narrative text features significantly improve model performance compared with structured variables only.	Model comparison using ROC-AUC, F1-score, and PR-AUC
RQ4: Are company-level and geographic differences significantly associated with complaint resolution outcomes after controlling for product and issue?	Company and geography are not statistically significantly associated with complaint resolution outcomes after controlling for other factors.	Company and geography are statistically significantly associated with complaint resolution outcomes after controlling for other factors.	Logistic regression with controls; feature importance; SHAP

Note. The hypotheses are directional where theory suggests a relationship, but the statistical tests will be interpreted using a significance level of $\alpha = .05$ unless otherwise justified.

Sample size calculation

The CFPB dataset is large enough to support the proposed analyses. However, the synopsis requires a sample-size justification for each research question. The sample-size plan below uses conservative assumptions. For chi-square tests, the study assumes a small effect size of Cohen's $w = .10$, $\alpha = .05$, and power = .80. For model-comparison evaluation, the study uses a confidence-interval approach for classification accuracy with $p = .50$ and margin of error $e = .02$, which gives a required test set of approximately 2,401 records. With an 80/20 train-test split, this translates to approximately 12,005 total records. For company-level modeling with many

predictors and potentially imbalanced outcomes, the study uses an events-per-variable rule as a practical safeguard. A minimum of 50,000 records will be used to ensure enough observations for minority outcome classes and stable model validation. The RQ4 safeguard is intentionally conservative because company-level controls, sparse product and issue categories, and rare resolution outcomes require a larger validation base than simple bivariate testing.

Table 2

Sample Size Calculation by Research Question

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Power analysis for chi-square test of independence	$\alpha = .05$, Power = .80, Effect size $w = .10$, $df \approx 4$	1,194
RQ2	Power analysis for chi-square / multinomial association screen	$\alpha = .05$, Power = .80, Effect size $w = .10$, $df \approx 18$	2,014
RQ3	Confidence interval for classification evaluation	$Z = 1.96$, $p = .50$, $e = .02$; 20% holdout test set	12,005 total records
RQ4	Events-per-variable safeguard for logistic/company controls	Approx. 50 effective predictors; 20 events per variable; low minority class assumption	50,000

To ensure adequate statistical power across all analyses, the final minimum sample size selected for the study will be 50,000 complaint records. If the filtered CFPB retail banking dataset contains more than 50,000 eligible records, the project will use the full available dataset or a stratified sample that preserves product, issue, year, and outcome distributions. A separate subset of records with published complaint narratives will be used for the NLP model comparison in RQ3.

Data Description

The primary data source will be the CFPB Consumer Complaint Database. The dataset is publicly available through the CFPB website and through the CFPB open data API. The CFPB

describes the database as a collection of complaints about consumer financial products and services that are sent to companies for response. Complaints are published after the company responds, confirms a commercial relationship with the consumer, or after 15 days, whichever comes first. The database generally updates daily (Data.gov, 2025; CFPB Open Tech, 2025).

The data will be downloaded from the official CFPB source rather than from a secondary Kaggle copy. The full dataset can be downloaded as CSV or JSON, and filtered subsets can be exported from the CFPB database interface. The API can also be used for reproducible extraction. The proposed GitHub repository will include the extraction script, data dictionary, transformation code, model code, validation outputs, and final tables. If the full raw dataset is too large for GitHub storage, the repository will include an extraction script and a data-access readme that reproduces the dataset from the official CFPB source.

Official data URLs:

- CFPB Consumer Complaint Database: <https://www.consumerfinance.gov/data-research/consumer-complaints/>
- CFPB field reference: <https://cfpb.github.io/api/ccdb/fields.html>
- CFPB search API documentation: <https://cfpb.github.io/ccdb5-api/>
- Data.gov catalog record: <https://catalog.data.gov/dataset/consumer-complaint-database>

Data dictionary (mandatory)

Table 3 presents the proposed working data dictionary. The exact categories will be confirmed after data extraction because CFPB product and issue taxonomies have changed over time. All transformations will be documented in the GitHub repository.

Table 3*Working Data Dictionary*

Field	Description	Data Type	Role in Study	Treatment / Notes
date_received	Date the CFPB received the complaint	Date	Control / trend variable	Convert to year, quarter, month; exclude incomplete most recent period if needed
product	Financial product selected by consumer	Categorical	Predictor; RQ1	Filter to retail banking-related products; one-hot encode
sub_product	More detailed product category	Categorical	Predictor	Use where coverage is sufficient; combine rare categories
issue	Main complaint issue selected by consumer	Categorical	Predictor; RQ2	Standardize issue taxonomy; one-hot encode or target encode
sub_issue	Detailed issue category	Categorical	Predictor	Use if not too sparse; combine rare categories
consumer_complaint_narrative	Consumer-submitted description where consent was given and personal information removed	Text	NLP predictor; RQ3	Clean text; TF-IDF; topic features; text length
company_public_response	Optional public-facing company response	Categorical/text	Post-response context; leakage-sensitive	Descriptive use only. Exclude from intake-stage predictive features.
company	Company named in complaint	Categorical	Predictor; RQ4	Limit to companies with sufficient volume; consider market-size limitation
state	Consumer state	Categorical	Predictor; RQ4	Use state controls; interpret with population caveat
zip_code	Published ZIP or partial ZIP when available	Text	Optional geographic feature	Use only aggregated level; avoid overfitting and privacy concerns
submitted_via	Complaint submission channel	Categorical	Predictor	Web, phone, referral, postal mail, or other
date_sent_to_company	Date CFPB sent complaint to company	Date	Process variable; leakage-sensitive	Use for process analysis unless known at prediction time. Exclude from intake-stage model.
company_response_to_consumer	Company response category	Categorical	Primary target / outcome	Collapse into analysis-ready classes if needed
timely_response	Whether company responded in timely manner	Binary	Primary target / outcome	Yes/No; model as binary classification
consumer_disputed	Whether consumer disputed response	Categorical	Historical/optional outcome	Availability may vary by period; use only if suitable
complaint_id	Unique complaint identifier	Identifier	Record key	Remove from model predictors

GitHub, data, and code access plan

The project will provide open access to the reproducible code and project artifacts through GitHub. Because the full CFPB dataset can be large, the repository will either include a cleaned sample dataset suitable for review or provide a reproducible script that downloads the same filtered dataset from the official CFPB API. The repository will include a README file explaining the project, data source, environment setup, folder structure, and instructions for rerunning the analysis.

Proposed repository tree:

```
retail-complaint-resolution-ml/
├── README.md
├── data/
│   ├── raw/                # Raw Data / API extract or link instructions
│   ├── processed/          # cleaned analysis-ready data
│   └── data_dictionary.csv
├── notebooks/
│   ├── 01_data_extraction.ipynb
│   ├── 02_exploratory_analysis.ipynb
│   ├── 03_hypothesis_tests.ipynb
│   ├── 04_model_training.ipynb
│   └── 05_model_interpretation.ipynb
├── src/
│   ├── preprocessing.py
│   ├── features.py
│   ├── modeling.py
│   └── evaluation.py
├── outputs/
│   ├── figures/
│   ├── tables/
│   └── model_metrics.csv
└── requirements.txt
```

Analytic approach

The analytic approach will proceed in five stages. First, the data will be extracted from the CFPB website or API and filtered to the selected date range and retail banking product categories. Second, the dataset will be cleaned by handling missing values, standardizing categorical values, removing duplicate complaint identifiers, creating time variables, and collapsing rare categories where necessary. Third, descriptive statistics and visualizations will

summarize complaint volume, products, issues, response types, channel patterns, and trends by year.

Fourth, hypothesis testing will be conducted for each research question. Categorical relationships will be tested with chi-square tests, and regression models will be used to estimate the effect of product type, issue category, submission channel, company, geography, and time controls on complaint outcomes. The study will report statistical significance, effect sizes, confidence intervals, and practical interpretation. Statistical significance alone will not be treated as business importance.

Fifth, machine learning models will be trained and compared. Logistic regression will serve as the baseline because it is interpretable and aligns with hypothesis testing. Decision tree and random forest models will provide nonlinear comparison. Gradient boosting, such as XGBoost, may be used if it improves performance and can be explained responsibly. Narrative models will use TF-IDF and possibly topic modeling. Model interpretation will rely on feature importance and SHAP-style explanations, while final recommendations will focus on how a bank could use the model as a decision-support tool rather than an automated decision maker.

Methodological refinements: class imbalance, NLP feature engineering, and data leakage prevention

Because complaint-resolution outcomes may be imbalanced, the study will explicitly test imbalance-mitigation methods during model training. Outcomes such as untimely response, monetary relief, or non-monetary relief may occur less frequently than routine closure outcomes. Therefore, the project will compare baseline models with data-level and algorithm-level imbalance treatments. Data-level methods will include Synthetic Minority Oversampling Technique (SMOTE) and targeted undersampling of majority-class records. Algorithm-level

methods will include class-weighted logistic regression, class-weighted random forest, and gradient boosting models with imbalance-sensitive parameters. Model performance will be evaluated using precision, recall, F1-score, ROC-AUC, and PR-AUC, with PR-AUC emphasized when the positive outcome class is rare.

For the natural language processing component, the study will extract structured features from consumer complaint narratives where narratives are available. Text preprocessing will include lowercasing, punctuation removal, stop-word removal, tokenization, and optional lemmatization. The main text representation method will be term frequency-inverse document frequency (TF-IDF). Additional features will include complaint length, selected keyword indicators, and topic features generated through Latent Dirichlet Allocation (LDA). LDA will be used to identify broader complaint themes that may not be fully represented by structured product and issue categories.

To prevent predictive data leakage, the study will exclude fields that would not be available at the intended prediction point. The company response field and company public response field will not be used as predictors when predicting response outcomes. Similarly, computed lag variables, including any lag between date received and date sent to company, will only be used if they are realistically available at the point of prediction. If they are not available at complaint intake, they will be reserved for descriptive process analysis rather than predictive modeling. This ensures the model functions as a true early-warning and prioritization tool rather than a retrospective explanation of completed complaint handling.

Solution to RQ1-4

RQ1: Product type and timely response

RQ1 will test whether the complaint product type is associated with a timely response. The initial test will be a chi-square test of independence between product category and timely response. A binary logistic regression model will then estimate the probability of a timely response while controlling for year, channel, company, and issue category. The proposed output will include odds ratios showing which product categories are associated with higher or lower response timeliness.

RQ2: Issue category and company response outcome

RQ2 will examine whether complaint issue categories predict company response outcomes. The outcome may be grouped into response categories such as closed with explanation, closed with monetary relief, closed with non-monetary relief, and other/administrative outcomes. The study will use multinomial logistic regression for statistical inference and a random forest model for predictive comparison. The expected business output is a ranked list of issue categories most associated with relief-based or unfavorable response outcomes.

RQ3: Added value of complaint narratives

RQ3 will compare a structured-field-only model with a structured-plus-narrative model. Narrative features will be created from published complaint narratives using TF-IDF, text length, selected keywords, and topic features. The null hypothesis will be rejected only if the narrative-enhanced model improves evaluation metrics meaningfully on a holdout test set. The study will avoid using post-outcome fields that create data leakage. It will also acknowledge that narratives

are available only when consumers consent, so narrative-based results may not generalize to all complaints.

For the narrative-based model, the study will use TF-IDF, complaint length, selected keyword indicators, and LDA topic probabilities. The structured-only model and the structured-plus-narrative model will be compared on the same holdout test set using ROC-AUC, PR-AUC, precision, recall, and F1-score. The narrative-enhanced model will be considered useful only if it improves predictive performance and produces interpretable features that can support practical complaint triage.

RQ4: Company and geographic differences

RQ4 will test whether company-level and geographic factors remain significant after controlling for product and issue. Because company size and market share are not fully included in the CFPB dataset, company effects will be interpreted carefully. The model will include company and state as controls or encoded features for prediction. Results will be framed as complaint-resolution risk signals rather than as definitive company performance rankings. Where feasible, the study may supplement state-level findings with public population data for contextual interpretation.

Evaluation metrics with formulae and calculations

The project will report both statistical and machine learning evaluation metrics. For hypothesis tests, the study will report p-values, confidence intervals, effect sizes, and odds ratios. For classification models, the study will report confusion matrices, accuracy, precision, recall, F1-score, ROC-AUC, and, when the outcome is imbalanced, precision-recall AUC. Because complaint outcomes may be imbalanced, accuracy will not be used alone.

Table 4*Model Evaluation Metrics and Formulas*

Metric	Formula	Use in Project
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correct classification rate; used cautiously under imbalance
Precision	$TP / (TP + FP)$	How many predicted high-risk complaints were truly high risk
Recall / Sensitivity	$TP / (TP + FN)$	How many actual high-risk complaints the model detected
F1-score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Balance between precision and recall
Specificity	$TN / (TN + FP)$	How well the model identifies non-high-risk complaints
ROC-AUC	Area under ROC curve	Threshold-independent discrimination ability
PR-AUC	Area under precision-recall curve	Preferred when the positive class is rare
Log loss	$-1/N \sum [y \log(p) + (1-y) \log(1-p)]$	Quality of predicted probabilities

Recommendation and application

The proposed application is a complaint-resolution risk model that supports prioritization and root-cause analysis. At complaint intake or early review, the model could estimate the probability that a complaint will experience delayed response, require relief, or fall into a high-risk outcome category. Complaint teams could use this information to route cases to the right queue, assign experienced handlers, monitor emerging issue clusters, and escalate recurring product problems to operations and digital-channel owners.

The model should not be used to deny service, minimize legitimate complaints, or automatically determine customer remedies. Instead, it should operate as a human-in-the-loop decision-support tool. A bank could combine model outputs with complaint policy, regulatory requirements, customer vulnerability indicators, and service-level agreements. The project will recommend governance controls such as model monitoring, bias review, explainability standards, audit logs, periodic retraining, and clear accountability for human decisions.

The expected business recommendations will include: (a) priority rules for complaint types with high predicted resolution risk, (b) dashboard indicators for product and issue categories with rising risk, (c) suggested service-level controls by channel and product, (d) narrative-based themes for root-cause analysis, and (e) responsible AI controls for transparent and fair complaint triage.

Identify the target user

The primary target users are retail banking complaint management leaders, customer experience leaders, operational-risk teams, compliance teams, and digital banking product owners. Secondary users include data analytics teams, audit teams, and senior executives responsible for customer conduct and service quality. Regulators and consumer-protection analysts may also benefit from the study because it demonstrates how public complaint data can be transformed into early-warning indicators while preserving interpretability and caution about representativeness.

Limitations and ethical considerations

The study has several limitations. First, CFPB complaint data are not a representative sample of all banking customers or all complaint experiences. Second, complaint narratives are available only when consumers consent, creating possible selection bias. Third, company-level comparisons may be affected by differences in company size, market share, product mix, and customer base. Fourth, the dataset captures published complaint records, not full internal bank investigation files. Fifth, variables such as customer lifetime value, internal service history, vulnerability indicators, and actual case-handling notes are not available in the open dataset.

Ethically, the model must be designed for fair and transparent use. Complaint prediction should improve service quality, not reduce access to remedies or create unfair routing. The project will avoid using protected-class proxies where possible, will report limitations of geographic and text features, and will recommend human oversight. Model interpretation will be emphasized so that bank leaders understand which factors drive predictions and where model outputs require caution.

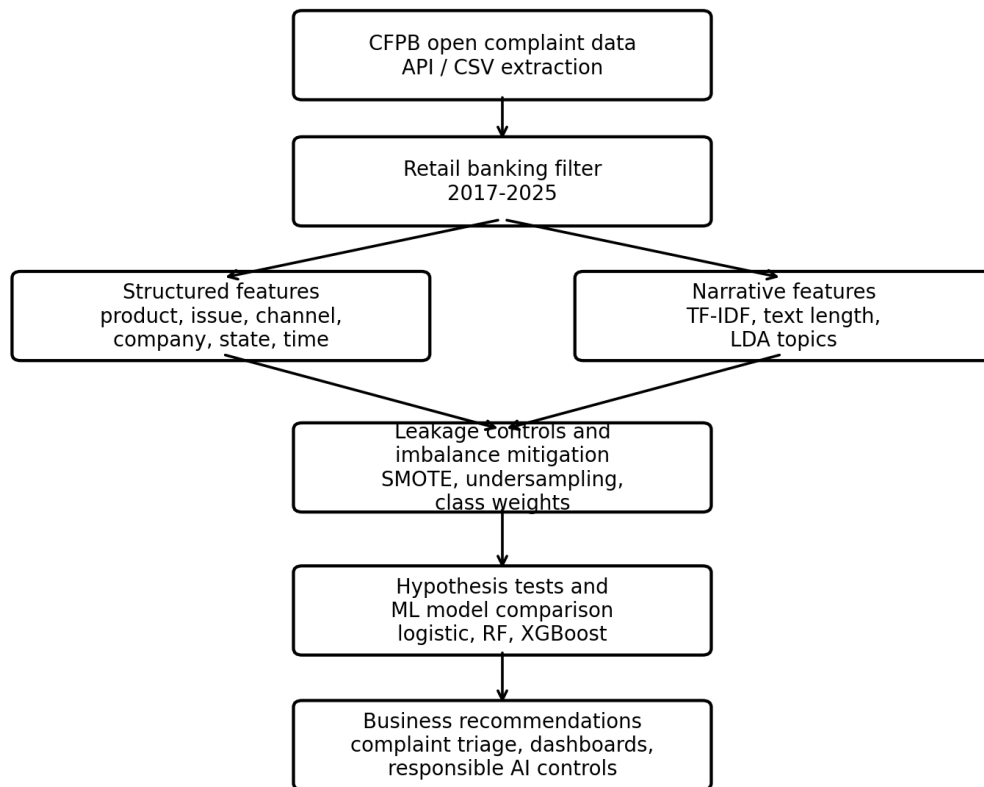


Figure 1. Proposed data analytics workflow with structured features, narrative features, leakage controls, imbalance mitigation, model comparison, and responsible AI application.

References and bibliography

- Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and machine learning: Limitations and opportunities. MIT Press.
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
<https://doi.org/10.1023/A:1010933404324>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321-357. <https://doi.org/10.1613/jair.953>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
- Consumer Financial Protection Bureau. (2025a). Consumer Complaint Database.
<https://www.consumerfinance.gov/data-research/consumer-complaints/>
- Consumer Financial Protection Bureau. (2025b). How we share complaint data.
<https://www.consumerfinance.gov/complaint/data-use/>
- CFPB Open Tech. (2025). Consumer Complaint Database API field reference.
<https://cfpb.github.io/api/ccdb/fields.html>
- CFPB Open Tech. (2025). Consumer Complaint Database search API.
<https://cfpb.github.io/ccdb5-api/>
- Data.gov. (2025). Consumer Complaint Database. <https://catalog.data.gov/dataset/consumer-complaint-database>

- Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In Proceedings of the 23rd International Conference on Machine Learning (pp. 233-240).
<https://doi.org/10.1145/1143844.1143874>
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Gao, P., Sun, N., Wang, X., Yang, C., & Zitakis, R. (2023). NLP-based detection of systematic anomalies among the narratives of consumer complaints. arXiv.
<https://arxiv.org/abs/2308.11138>
- Homburg, C., & Fürst, A. (2005). How organizational complaint handling drives customer loyalty: An analysis of the mechanistic and the organic approach. Journal of Marketing, 69(3), 95–114. <https://doi.org/10.1509/jmkg.69.3.95.66367>
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). Applied logistic regression (3rd ed.). Wiley.
- Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data, 6(4), 1-21. <https://doi.org/10.1145/2382577.2382579>
- Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. The Journal of Finance, 66(1), 35–65.
<https://doi.org/10.1111/j.1540-6261.2010.01625.x>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems, 30.

- Maxham, J. G., III, & Netemeyer, R. G. (2002). Modeling customer perceptions of complaint handling over time: The effects of perceived justice on satisfaction and intent. *Journal of Retailing*, 78(4), 239–252. [https://doi.org/10.1016/S0022-4359\(02\)00100-8](https://doi.org/10.1016/S0022-4359(02)00100-8)
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6), 1–35. <https://doi.org/10.1145/3457607>
- Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, 49(12), 1373–1379. [https://doi.org/10.1016/S0895-4356\(96\)00236-3](https://doi.org/10.1016/S0895-4356(96)00236-3)
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should I trust you?” Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144). <https://doi.org/10.1145/2939672.2939778>
- Tax, S. S., Brown, S. W., & Chandrashekar, M. (1998). Customer evaluations of service complaint experiences: Implications for relationship marketing. *Journal of Marketing*, 62(2), 60–76. <https://doi.org/10.1177/002224299806200205>
- Vaishnav, D., Neethinayagam, M., Khaire, A., & Woo, J. (2024). Predictive analysis of CFPB consumer complaints using machine learning. *arXiv*. <https://arxiv.org/abs/2407.06399>

Wang, Z., Zhu, S., & Chen, X. (2026). From complaint narratives to monetary relief: A hybrid machine learning framework for CFPB consumer complaints. arXiv.

<https://arxiv.org/abs/2606.22664>

Appendix A: Proposed project deliverables

The final capstone deliverables will include the written report, reproducible GitHub repository, cleaned data dictionary, hypothesis-testing notebook, machine learning notebook, evaluation tables, model interpretation outputs, and a management recommendation section. Screenshots of the CFPB data extraction interface, GitHub repository, and model output tables will be added in the final report after the analysis is completed.